

Advanced Machine Learning



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Administrative

- I've uploaded the solutions for seminar 2
- seminar 3 class today, 4 exercises
- seminar 3 also next Monday + Thursday
- seminar 4 starts after Easter holiday
- next week you get the assignment

Today's lecture: Overview

- Shattering
- VC-dimension

Shattering

How expressive is $\mathcal{H}$?

In our binary classification context, a hypothesis is a function $h: \mathcal{X} \rightarrow \{0, 1\}$

Hence the expressiveness of $\mathcal{H}$:

- is necessary a measure of how many functions $\mathcal{H}$ can express
- in the light of the No Free Lunch theorem, not only functions on $\mathcal{X}$, but also functions on (finite) subsets C of $\mathcal{X}$. We used this in the proof of NFL to construct a set of functions f_i and distributions $\mathcal{D}_i$ over $C \times \{0, 1\}$ such that there exists a distribution on which a learning algorithm will fail but $L_{\mathcal{D}_i}(f_i) = 0$.

Definition (restriction of $\mathcal{H}$ to C)

Let $\mathcal{H}$ be a set of hypothesis, i.e., set of functions from $\mathcal{X}$ to $\{0, 1\}$, and let C be a (finite) subset of $\mathcal{X}$, $C = \{c_1, c_2, \dots, c_m\}$. The restriction of $\mathcal{H}$ to C , denoted by $\mathcal{H}_C$, is the set of functions from C to $\{0, 1\}$ that can be derived from $\mathcal{H}$. That is:

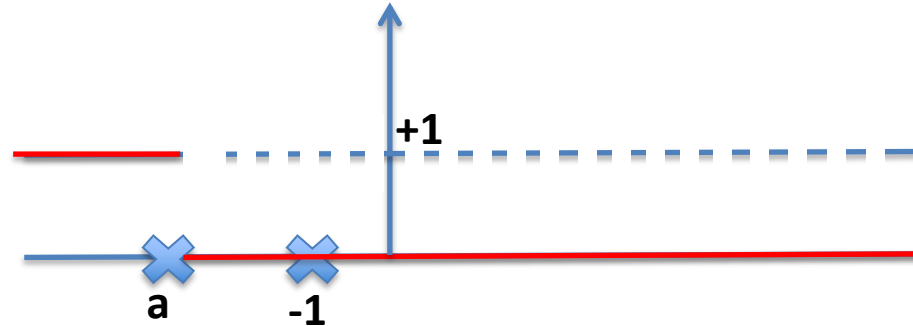
$$\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$$

Example: restriction of $\mathcal{H}$ to C

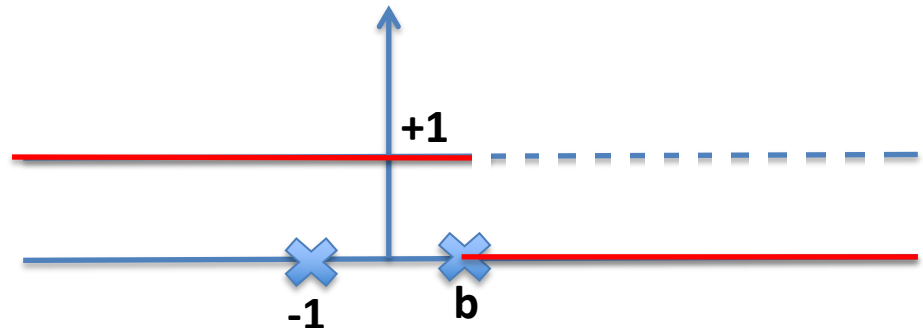
Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line
 $\mathcal{H}_{\text{thresholds}} = \{h_a: \mathbb{R} \rightarrow \{0, 1\}, h_a(x) = \mathbf{1}_{[x < a]}, a \in \mathbb{R}\}, |\mathcal{H}_{\text{thresholds}}| = \infty.$

Consider $C = \{-1\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\} = \{h: \{-1\} \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has 2 elements h_a, h_b where:

$$h_a(-1) = 0, a \leq -1$$



$$h_b(-1) = 1, b > -1$$

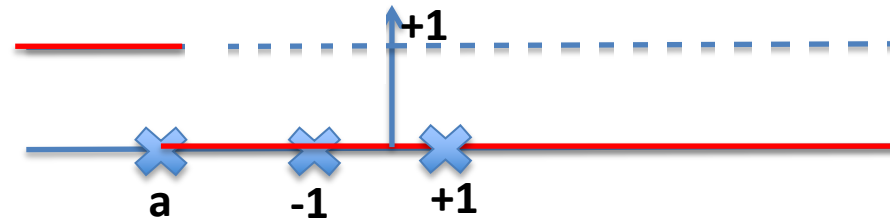


Example: restriction of $\mathcal{H}$ to C

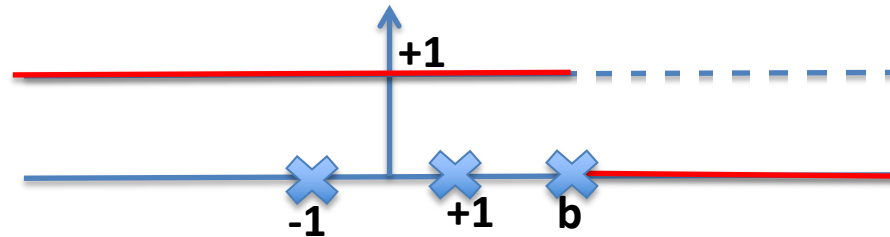
Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line
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Consider $C = \{-1, 1\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\} = \{h: \{-1, 1\} \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has 3 elements h_a, h_b, h_c where:

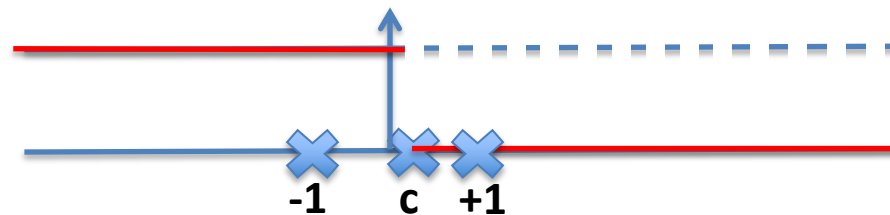
$$h_a(-1) = 0, h_a(1) = 0, a \leq -1$$



$$h_b(-1) = 1, h_b(1) = 1, b > 1$$



$$h_c(-1) = 1, h_c(1) = 0, -1 < c \leq 1$$



Example: restriction of $\mathcal{H}$ to C

Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line
 $\mathcal{H}_{\text{thresholds}} = \{h_a: \mathbb{R} \rightarrow \{0, 1\}, h_a(x) = \mathbf{1}_{[x < a]}, a \in \mathbb{R}\}, |\mathcal{H}_{\text{thresholds}}| = \infty.$

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$$h_a(-1) = 0, h_a(1) = 0, a \leq -1$$

$$h_b(-1) = 1, h_b(1) = 1, b > 1$$

$$h_c(-1) = 1, h_c(1) = 0, -1 < c \leq 1$$

There is no function h_d in $\mathcal{H}_C$ such that $h_d(-1) = 0$ and $h_d(1) = 1$ (as we work with the threshold functions)

Alternative view of functions

There is equivalence between functions from $\mathcal{X}$ to $\{0,1\}$ and subsets C of $\mathcal{X}$

- given $h: \mathcal{X} \rightarrow \{0, 1\}$, we can define $C = \{x \in \mathcal{X} \mid \text{such that } h(x) = 1\}$
- given C subset of $\mathcal{X}$, we can define $h: \mathcal{X} \rightarrow \{0,1\}$, $h(x) = \mathbf{1}_C(x)$ indicator function of C

Can represent a subset C of $\mathcal{X}$ as a vector, put 1 for element in C , 0 otherwise.

Can represent each function from C to $\{0,1\}$ as a vector in $\{0,1\}^{|C|}$.

The restriction of $\mathcal{H}$ to C , denoted by $\mathcal{H}_C$, is the set of functions from $C = \{c_1, c_2, \dots, c_m\}$ to $\{0, 1\}$ that can be derived from $\mathcal{H}$. It can be also seen as the set of all possible vectors that can be generated by $h \in \mathcal{H}$ with elements from C . That is:

$$\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\} = \{(h(c_1), h(c_2), \dots, h(c_m)) \mid h \in \mathcal{H}\}$$

Consider $C = \{-1, 1\}$. Then $\mathcal{H}_{\text{thresholds}/C} = \mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\} = \{h: \{-1, 1\} \rightarrow \{0, 1\} \mid h \in \mathcal{H}\} = \{(0,0), (1,0), (1,1)\}$. **The vector (0,1) is not realizable by $\mathcal{H}_C$.**

Shattering

Definition (Shattering)

A hypothesis class $\mathcal{H}$ *shatters* a finite set C of $\mathcal{X}$, if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0, 1\}$. That is $|\mathcal{H}_C| = 2^{|C|}$.

Examples:

Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line.

Consider $C = \{c_1\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has two elements $\{h_a, h_b\}$ with $a \leq c_1$ and $b > c_1$ so $\mathcal{H}$ shatters C . $\mathcal{H}_C = \{(0), (1)\}$, $|\mathcal{H}_C| = 2^{|C|} = 2^1$

Consider $C = \{c_1, c_2 \mid c_1 < c_2\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has three elements, there is no function that realizes the labeling $(0, 1)$ and so $\mathcal{H}$ does not shatter C .

Alternative view of Shattering

Definition (Shattering)

A hypothesis class $\mathcal{H}$ *shatters* a finite set C of $\mathcal{X}$, if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0, 1\}$. That is $|\mathcal{H}_C| = 2^{|C|}$.

Similar Definition (Shattering)

Let $\mathcal{H}$ be a collection of subsets of $\mathcal{X}$ and C a finite subset of $\mathcal{X}$.

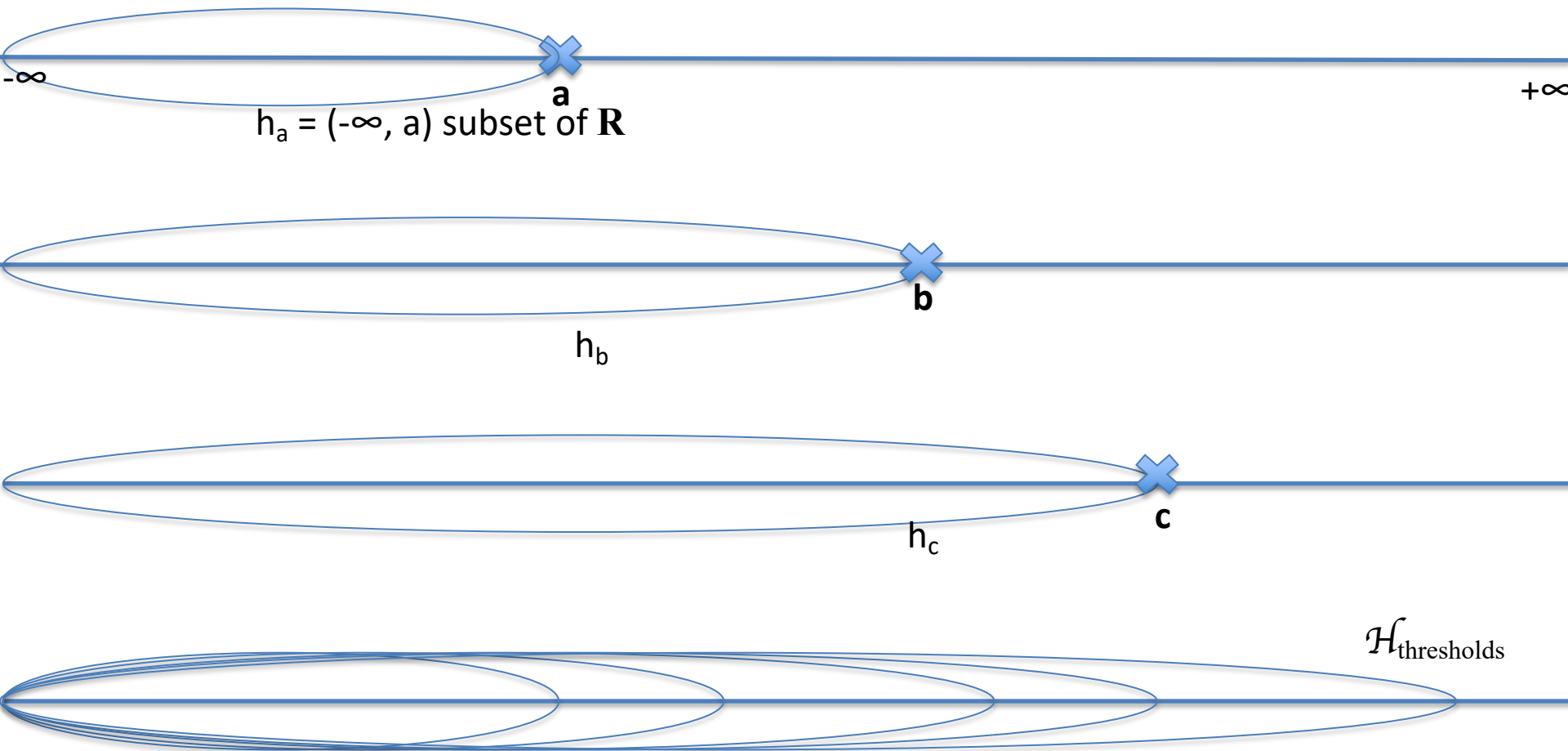
$\mathcal{H}$ shatters C if for every subset B of C there exist some subset $h_B \in \mathcal{H}$ such that $B = h_B \cap C$

In other words, using the elements of $\mathcal{H}$, we can cut C in every possible way.

Shattering – graphical representation

Examples:

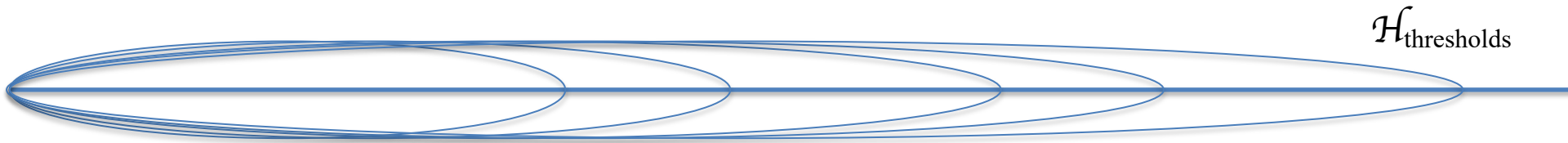
Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line.



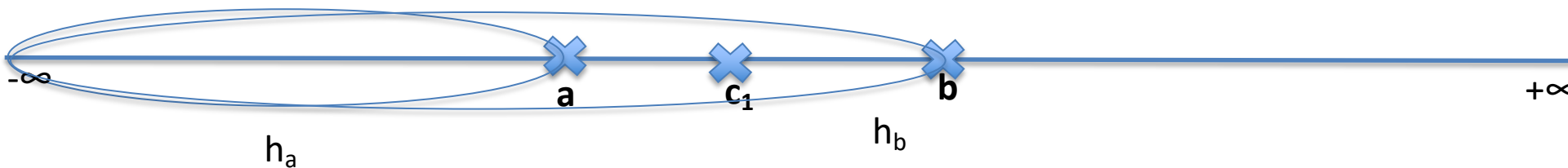
Shattering – graphical representation

Examples:

Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line.



Consider $C = \{c_1\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has two elements $\{h_a, h_b\}$ with $a \leq c_1$ and $b > c_1$ so $\mathcal{H}$ shatters C . $\mathcal{H}_C = \{(0), (1)\}$, $|\mathcal{H}_C| = 2^{|C|} = 2^1$



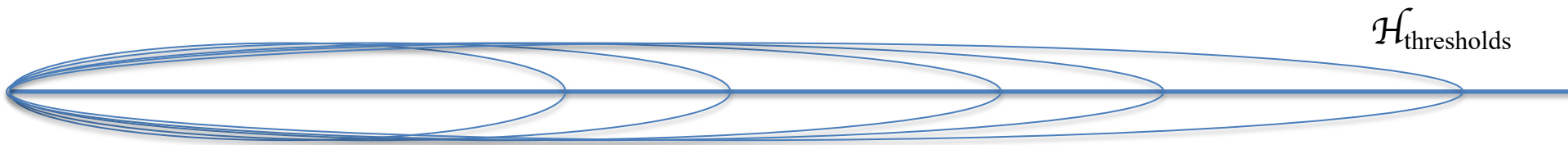
h_a generates label 0 (corresponding to the empty set $= C \cap h_a$)

h_b generates label 1 (corresponding to subset $C = C \cap h_b$)

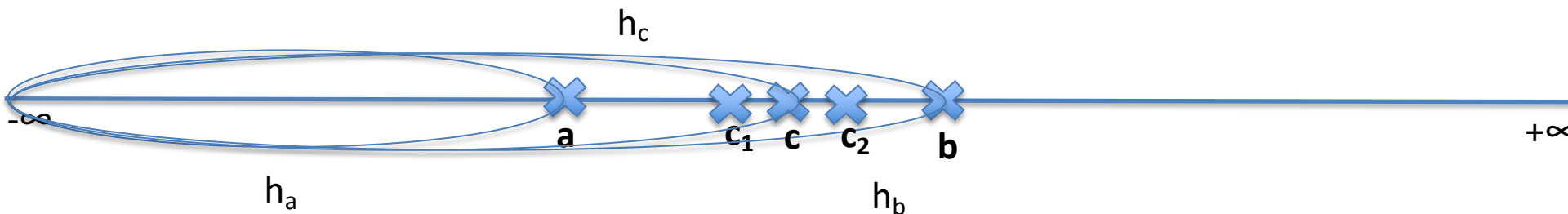
Shattering – graphical representation

Examples:

Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line.



Consider $C = \{c_1, c_2 \mid c_1 \leq c_2\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has at most three elements, there is no function that realizes the labeling (0,1) and so $\mathcal{H}$ does not shatter C .



h_a generates label (0,0) – corresponding to empty set

h_b generates label (1,1) – corresponding to $C = \{c_1, c_2\}$

h_c generates label (1,0) – corresponding to subset $\{c_1\}$

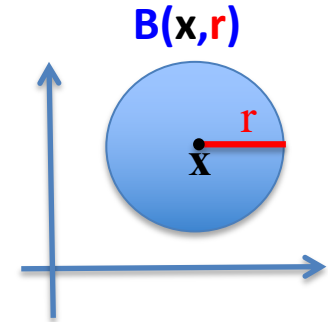
cannot generate the label (0,1)

Shattering – example $\mathcal{H}_{\text{balls}}$

Consider $\mathcal{H} = \mathcal{H}_{\text{balls}}$ be the set of all balls in $\mathbf{R}^2$:

$$\mathcal{H}_{\text{balls}} = \{B(\mathbf{x}, r), \mathbf{x} \in \mathbf{R}^2, r \geq 0\},$$

$$B(\mathbf{x}, r) = \{y \in \mathbf{R}^2 \mid \|y - \mathbf{x}\|_2 \leq r\}$$

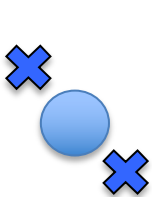


Can also view $\mathcal{H}_{\text{balls}}$ as:

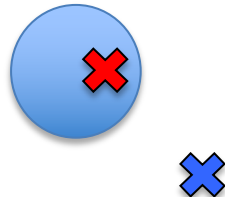
$$\mathcal{H}_{\text{balls}} = \{h_{\mathbf{x}, r}: \mathbf{R}^2 \rightarrow \{0, 1\}, h_{\mathbf{x}, r} = \mathbf{1}_{B(\mathbf{x}, r)}, \mathbf{x} \in \mathbf{R}^2, r \geq 0\}$$

Is there a set A in $\mathbf{R}^2$ of size 2 shattered by $\mathcal{H}_{\text{balls}}$?

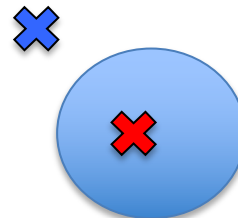
Any set A of two distinct points in $\mathbf{R}^2$ is shattered by $\mathcal{H}_{\text{balls}}$



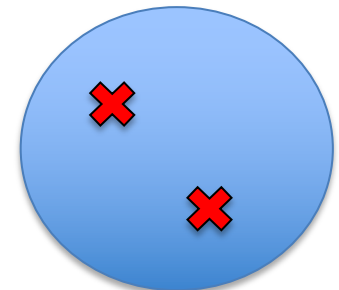
label (0,0)



label (1,0)



label (0,1)

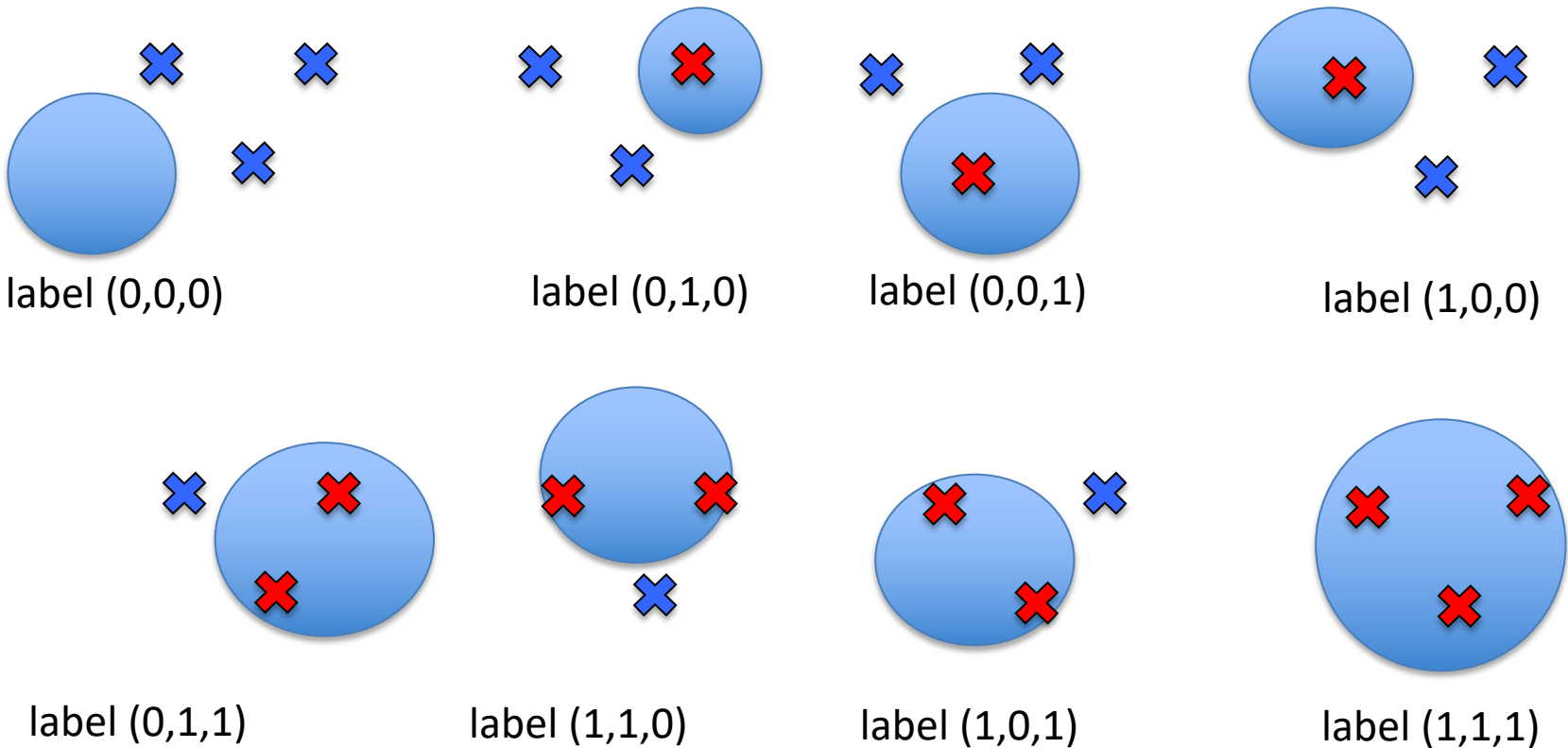


label (1,1)

Shattering – example $\mathcal{H}_{\text{balls}}$

Is there a set A in $\mathbf{R}^2$ of size 3 shattered by $\mathcal{H}_{\text{balls}}$?

Any set A of three distinct points in $\mathbf{R}^2$ that are not collinear is shattered by $\mathcal{H}_{\text{balls}}$



Shattering – example $\mathcal{H}_{\text{balls}}$

Is there a set A in $\mathbf{R}^2$ of size 3 shattered by $\mathcal{H}_{\text{balls}}$?

Any set A of three distinct points in $\mathbf{R}^2$ that are collinear is not shattered by $\mathcal{H}_{\text{balls}}$



cannot realize the label $(1, 0, 1)$

What are the conditions for which a set A in $\mathbf{R}^2$ of size 4 is shattered by $\mathcal{H}_{\text{balls}}$?

No Free lunches - revisited

- from the proof of the No Free Lunch theorem, we saw that we can create an adversarial distribution if $\mathcal{H}$ shatters a too large class.
- let $\mathcal{H}$ be a hypothesis class of functions $h: \mathcal{X} \rightarrow \{0, 1\}$ and S a training set of size m . If there exists a set $C \subseteq \mathcal{X}$ of size $2m$ that is shattered by $\mathcal{H}$, then for any learning algorithms A who receives S there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$ such that:
 - there exists a function $f: \mathcal{X} \rightarrow \{0, 1\}$ with $L_{\mathcal{D}}(f) = 0$
 - with probability of at least $1/7$ over the choice of $S \sim \mathcal{D}^m$ we have that $L_{\mathcal{D}}(A(S)) \geq 1/8$
- the labels of the m instances give us no information about the labels of the rest of the instances in C – every possible labeling of the rest of the instances can be explained by some hypothesis in $\mathcal{H}$.
- *“If someone can explain every phenomenon, his explanations are worthless”*
- a model that can fit any labeling has zero predictive power because it imposes no structure on the data. Shattering is good, but don't shatter too much.

VC-dimension

The VC-dimension

Definition (VC-dimension)

The VC - dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the maximal size of a set $C \subseteq \mathcal{X}$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size we say that $\mathcal{H}$ has infinite VC-dimension.

Theorem

Let $\mathcal{H}$ be a class of infinite VC-dimension. Then, $\mathcal{H}$ is not PAC learnable.

Proof: Since $\mathcal{H}$ has an infinite VC-dimension, for any training set S of size m , there exists a shattered set of size $2m$, and the claim follows for the No Free Lunch theorem.

We will see in the next lecture that the converse is also true: *a finite VC-dimension guarantees learnability. Hence, the VC-dimension characterizes PAC learnability. VC-dimension is a combinatorial measure, does not imply computing probabilities.*

Determining the VC-dimension of $\mathcal{H}$

Definition (VC-dimension)

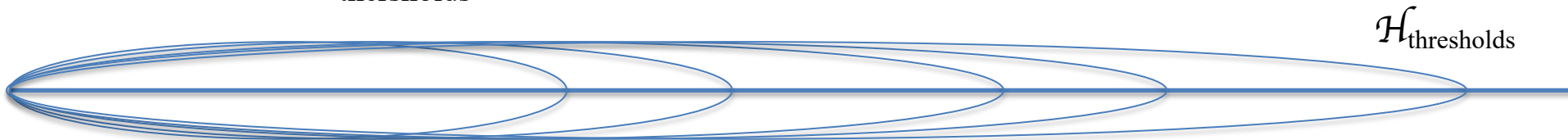
The VC - dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the maximal size of a set $C \subseteq X$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size we say that $\mathcal{H}$ has infinite VC-dimension.

In order to show that the VC-dimension of a hypothesis class $\mathcal{H}$ is d , we need to show that:

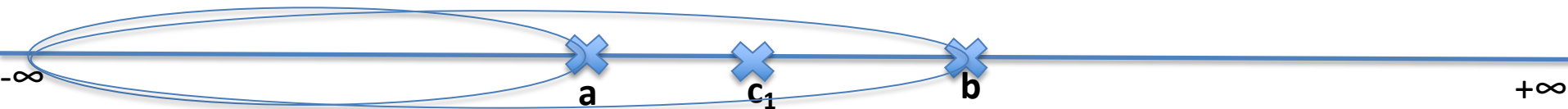
- 1. There exists a set C of size d that is shattered by $\mathcal{H}$. ($\text{VCdim}(\mathcal{H}) \geq d$)**
- 2. Every set C of size $d + 1$ is not shattered by $\mathcal{H}$. ($\text{VCdim}(\mathcal{H}) < d+1$)**

VCdim($\mathcal{H}_{\text{thresholds}}$)

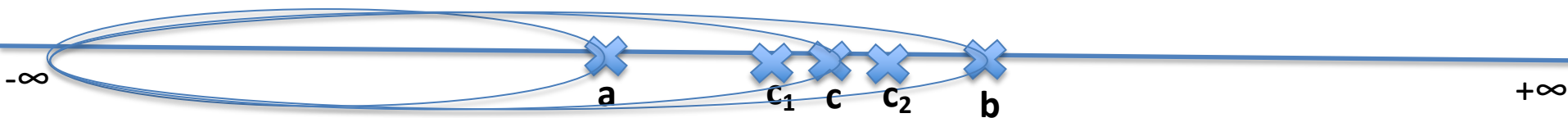
Consider $\mathcal{H} = \mathcal{H}_{\text{thresholds}}$ be the set of threshold functions over the real line.



Consider $C = \{c_1\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has two elements $\{h_a, h_b\}$ with $a \leq c_1$ and $b > c_1$ so $\mathcal{H}$ shatters C . $\mathcal{H}_C = \{(0), (1)\}$, $|\mathcal{H}_C| = 2^{|C|} = 2^1$



Consider $C = \{c_1, c_2 \mid c_1 \leq c_2\}$. Then $\mathcal{H}_C = \{h: C \rightarrow \{0, 1\} \mid h \in \mathcal{H}\}$ has at most three elements, there is no function that realizes the labeling (0,1) and so $\mathcal{H}$ does not shatter C .



So, **VCdim($\mathcal{H}_{\text{thresholds}}$) = 1**

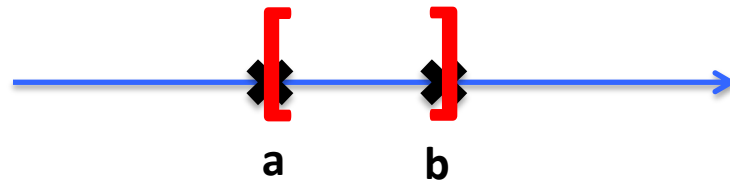
VCdim($\mathcal{H}_{\text{intervals}}$)

Consider $\mathcal{H} = \mathcal{H}_{\text{intervals}}$ be the set of intervals over the real line.

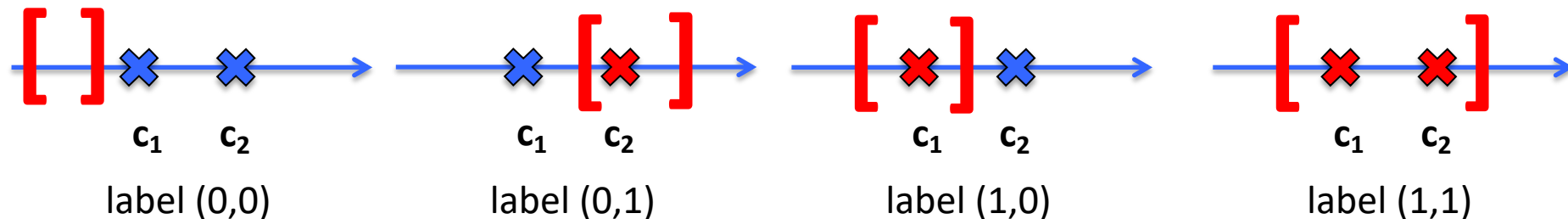
$$\mathcal{H}_{\text{intervals}} = \{[a,b] \mid a \leq b, a, b \in \mathbf{R}\}$$

Can also view $\mathcal{H}_{\text{intervals}}$ as:

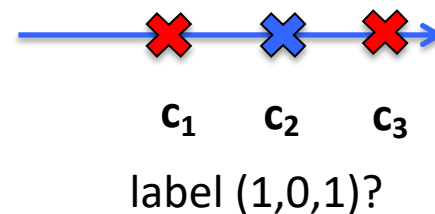
$$\mathcal{H}_{\text{intervals}} = \{h_{a,b}: \mathbf{R} \rightarrow \{0, 1\}, h_{a,b} = \mathbf{1}_{[a,b]}, a \leq b, a, b \in \mathbf{R}\}$$



$\mathcal{H}_{\text{intervals}}$ shatters any set A of two different points in $\mathbf{R}$.



$\mathcal{H}_{\text{intervals}}$ cannot shatter any set A of three points in $\mathbf{R}$.



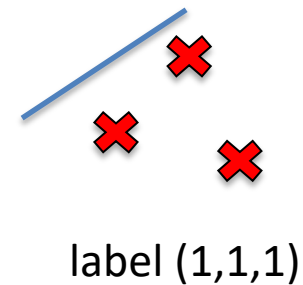
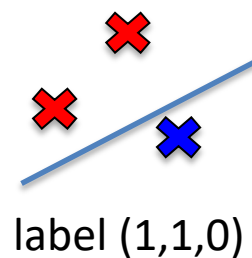
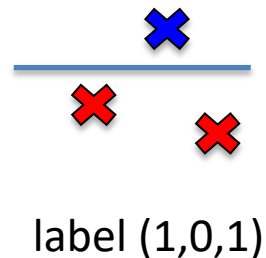
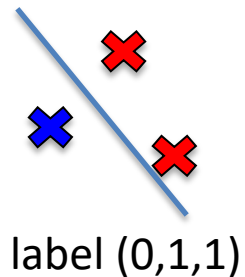
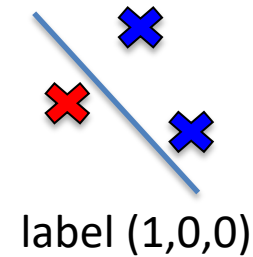
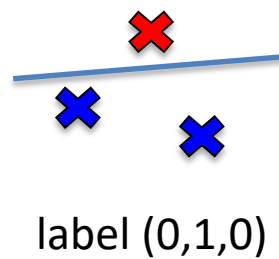
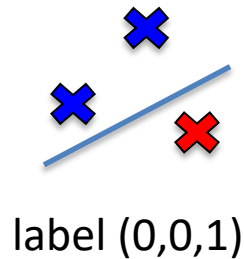
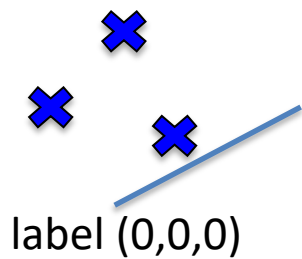
So, **VCdim($\mathcal{H}_{\text{intervals}}$) = 2**

VCdim($\mathcal{H}_{\text{lines}}$)

Consider $\mathcal{H} = \mathcal{H}_{\text{lines}}$ be the set of lines in $\mathbf{R}^2$.

$\mathcal{H}_{\text{lines}} = \{h_{a,b,c}: \mathbf{R}^2 \rightarrow \{0, 1\}, h_{a,b,c}((x,y)) = \mathbf{1}_{[ax+by+c>0]}((x,y)) , a, b, c \in \mathbf{R}\}$

$\mathcal{H}_{\text{lines}}$ shatters any set A of three non-collinear points in $\mathbf{R}^2$.



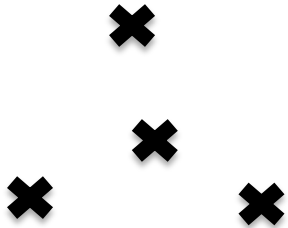
VCdim($\mathcal{H}_{\text{lines}}$)

Consider $\mathcal{H} = \mathcal{H}_{\text{lines}}$ be the set of lines in $\mathbf{R}^2$.

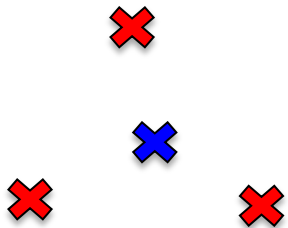
$\mathcal{H}_{\text{lines}} = \{h_{a,b,c}: \mathbf{R}^2 \rightarrow \{0, 1\}, h_{a,b,c}((x,y)) = \mathbf{1}_{[ax+by+c>0]}((x,y)) , a, b, c \in \mathbf{R}\}$

$\mathcal{H}_{\text{lines}}$ doesn't shatter any set A of four points in $\mathbf{R}^2$.

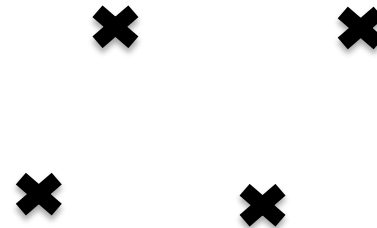
Case a: one point is interior to the convex hull of the other 3



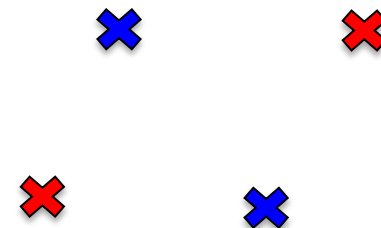
label (1,1,1,0) is un-realizable



Case b: no point is interior to the convex hull of the other 3



label (1,0,1,0) is un-realizable

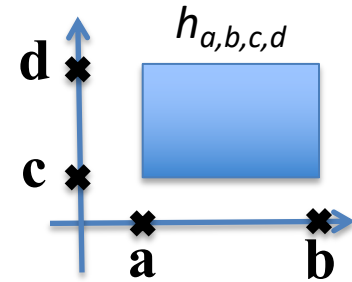


So, **VCdim($\mathcal{H}_{\text{lines}}$) = 3**

VCdim($\mathcal{H}_{\text{rec}}^2$)

Consider $\mathcal{H} = \mathcal{H}_{\text{rec}}^2$ be the set of axis aligned rectangles in the $\mathbf{R}^2$.

$$\mathcal{H}_{\text{rec}}^2 = \{[a,b] \times [c,d] \mid a \leq b, c \leq d, a, b, c, d \in \mathbf{R}\}$$

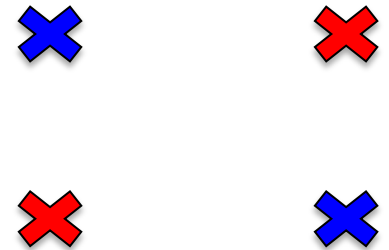


Can also view $\mathcal{H}_{\text{rec}}^2$ as:

$$\mathcal{H}_{\text{rec}}^2 = \{h_{a,b,c,d}: \mathbf{R}^2 \rightarrow \{0, 1\}, h_{a,b,c,d} = \mathbf{1}_{[a,b] \times [c,d]}, a \leq b, c \leq d, a, b, c, d \in \mathbf{R}\}$$

Does $\mathcal{H}_{\text{rec}}^2$ shatters any set A of four different points in $\mathbf{R}^2$?

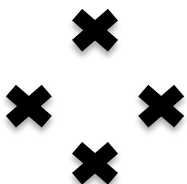
Take A the set of 4 vertices of a rectangle with axis aligned in $\mathbf{R}^2$. Then $\mathcal{H}_{\text{rec}}^2$ doesn't shatter A (the (0,1,0,1) labeling is not realizable).

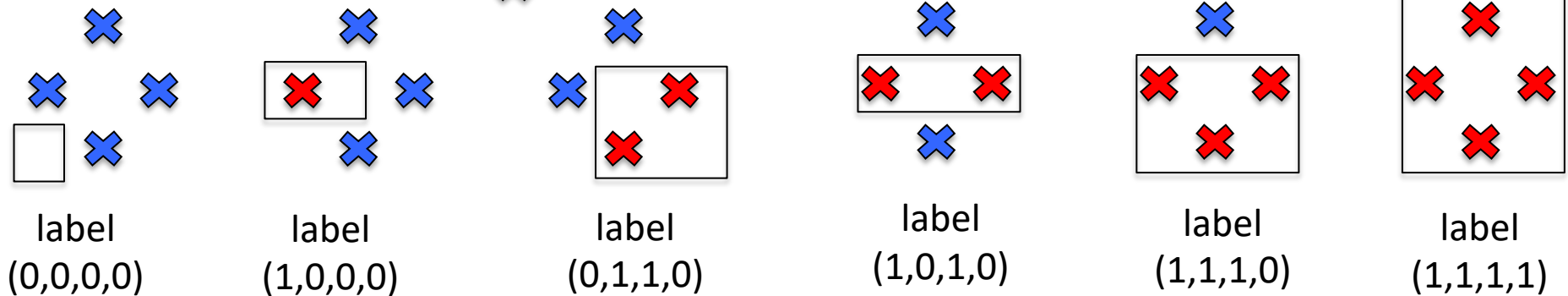


Does this mean that $\text{VCdim}(\mathcal{H}_{\text{rec}}^2) < 4$?

No! Either we show that all sets A of size 4 are not shattered by $\mathcal{H}_{\text{rec}}^2$ ($\text{VCdim}(\mathcal{H}_{\text{rec}}^2) < 4$) or find a set A of size 4 that is shattered ($\text{VCdim}(\mathcal{H}_{\text{rec}}^2) \geq 4$).

VCdim($\mathcal{H}_{\text{rec}}^2$)

Consider the set A:  Then A is shattered by $\mathcal{H}_{\text{rec}}^2$.

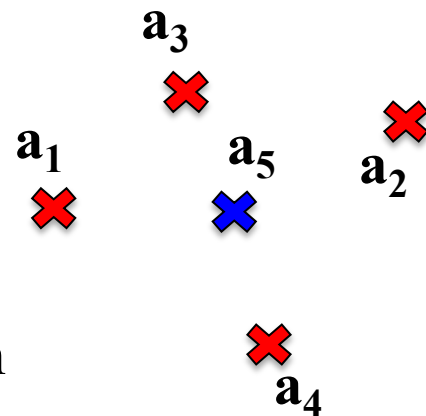


Can realize all 16 possible labels. So $\text{VCdim}(\mathcal{H}_{\text{rec}}^2) \geq 4$

Show now that all sets A of 5 points are not shattered by $\mathcal{H}_{\text{rec}}^2$.

A = {a₁, a₂, a₃, a₄, a₅}. Consider a₁ – the leftmost point (smaller x), a₂ – the rightmost point (larger x), a₃ – the lowest point (smaller y), a₄ – the highest point (larger y).

Then every rectangle containing a₁, a₂, a₃, a₄ will also contain a₅, so (1,1,1,1,0) is not realizable. So, **VCdim($\mathcal{H}_{\text{rec}}^2$) = 4**



Some basic properties of the $\text{VCdim}(\mathcal{H})$

1. $\text{VCdim}(\mathcal{H}) \leq \log_2 |\mathcal{H}|$
2. If $\mathcal{H}_1 \subseteq \mathcal{H}_2$ then $\text{VCdim}(\mathcal{H}_1) \leq \text{VCdim}(\mathcal{H}_2)$
3. If $\text{VCdim}(\mathcal{H}) = \infty$ then $\mathcal{H}$ is not PAC learnable

$\text{VCdim}(\mathcal{H})$ is the maximal size of a set $C \subseteq \mathcal{X}$ that can be shattered by $\mathcal{H}$.

$\text{VCdim}(\mathcal{H}_{\sin})$

$$\text{VCdim}(\mathcal{H}_{\text{thresholds}}) = 1, \text{VCdim}(\mathcal{H}_{\text{intervals}}) = 2, \text{VCdim}(\mathcal{H}_{\text{lines}}) = 3$$

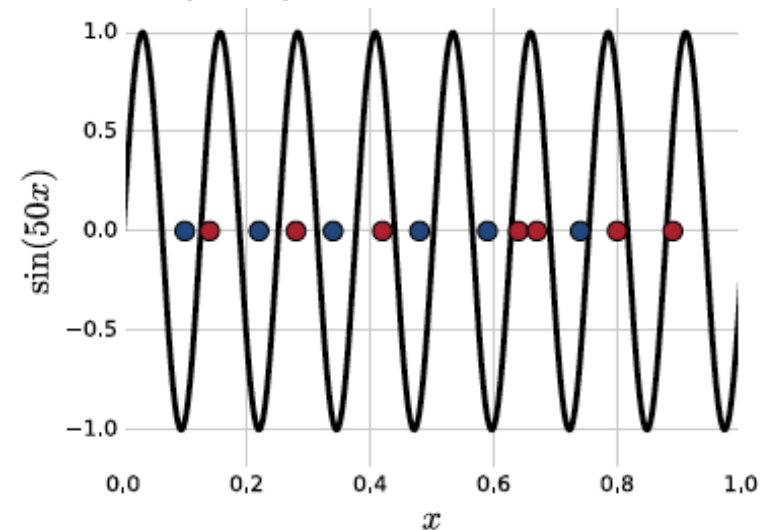
$$\text{VCdim}(\mathcal{H}_{\text{rec}}^2) = 4$$

Consider $\mathcal{H} = \mathcal{H}_{\sin}$ be the set of sin functions:

$$\mathcal{H}_{\sin} = \{h_{\theta}: \mathbf{R} \rightarrow \{0,1\} \mid h_{\theta}(x) = \lceil \sin(\theta x) \rceil, \theta \in \mathbf{R}\}, \quad \lceil -1 \rceil = 0$$

$$\text{VCdim}(\mathcal{H}_{\sin}) = ?$$

We will show that $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$



● Class 0

● Class 1

$\text{VCdim}(\mathcal{H}_{\sin})$

Lemma

Let $x \in (0, 1)$ and let $0.x_1x_2x_3\dots$ be the binary representation of x . Then, for any natural number m , provided that there exist $k \geq m$ such that $x_k = 1$, we have:

$$\left\lceil \sin(2^m \pi x) \right\rceil = 1 - x_m$$

Example of binary representation:

$$x = (0.x_1x_2x_3\dots)_2 = x_1 \times 2^{-1} + x_2 \times 2^{-2} + x_3 \times 2^{-3} + \dots$$

$$x = 0.75 = \frac{1}{2} + \frac{1}{4} = (0.110000\dots)_2$$

$$x = 0.3 = 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 0 \times 2^{-4} + 1 \times 2^{-5} + \dots = (0.01001\dots)_2$$

$\text{VCdim}(\mathcal{H}_{\sin})$

Lemma

Let $x \in (0, 1)$ and let $0.x_1x_2x_3\dots$ be the binary representation of x . Then, for any natural number m , provided that there exist $k \geq m$ such that $x_k = 1$, we have:

$$\left\lceil \sin(2^m \pi x) \right\rceil = 1 - x_m$$

Proof

$$\begin{aligned} \sin(2^m \pi x) &= \sin(2^m \pi (0.x_1x_2x_3\dots)) = \sin(2\pi * 2^{m-1} (0.x_1x_2x_3\dots)) = \\ &= \sin(2\pi * (x_1x_2x_3\dots x_{m-1}.x_m x_{m+1}\dots)) \text{ (left shift with } m-1 \text{ position)} \\ &= \sin(2\pi * (x_1x_2x_3\dots x_{m-1}.x_m x_{m+1}\dots) - 2\pi * (x_1x_2x_3\dots x_{m-1}.0)) \text{ (sin has period } 2\pi) \\ &= \sin(2\pi * (0.x_mx_{m+1}\dots)) \end{aligned}$$

Note that $0.x_mx_{m+1}\dots > 0$ as there exist $k \geq m$ such that $x_k = 1$

Case 1: $x_m = 0$, then $0 < 2\pi * (0.x_mx_{m+1}\dots) < 2\pi * 1/2 = \pi$. So $0 < \sin(2^m \pi x) < 1$, and from here it results that: $\left\lceil \sin(2^m \pi x) \right\rceil = 1 = 1 - 0 = 1 - x_m$

Case 2: $x_m = 1$, then $2\pi \geq 2\pi * (0.x_mx_{m+1}\dots) \geq 2\pi * 1/2 = \pi$. So $-1 \leq \sin(2^m \pi x) \leq 0$, and from here it results that: $\left\lceil \sin(2^m \pi x) \right\rceil = 0 = 1 - 1 = 1 - x_m$ (we consider $\left\lceil -1 \right\rceil = 0$)

VCdim($\mathcal{H}_{\text{sin}}$)

To prove $\text{VCdim}(\mathcal{H}_{\text{sin}}) = \infty$, we need to pick n points which are shattered by $\mathcal{H}_{\text{sin}}$, for any n . To do so, we construct n points $x_1, x_2, \dots, x_n \in [0, 1]$, such that the set of the m -th bits in the binary expansion, as m ranges from 1 to 2^n , ranges over all possible labelings of $x_1, x_2, \dots, x_n$.

$$\begin{array}{rcl} x_1 & = & 0.\textcolor{red}{0}0000\dots 11 \\ x_2 & = & 0.\textcolor{red}{0}0000\dots 11 \\ & \dots & \\ x_{n-1} & = & 00011\dots 11 \\ x_n & = & 0.\textcolor{red}{0}101\dots 01 \end{array}$$

$\textcolor{red}{m=1}$

For example, to give the labeling 1 for all instances, we just pick $\textcolor{red}{m=1}$:

$$h(x) = \lceil \sin(2^1 \pi x) \rceil = 1 - \textit{first_bit_of_binary_repres_of_}x$$

$$\mathcal{H}_{\text{sin}} = \{h_\theta: \mathbf{R} \rightarrow \{0,1\} \mid h_\theta(x) = \lceil \sin(\theta x) \rceil, \theta \in \mathbf{R}\},$$

which returns 1 - the first bit (column) in the binary expansion.

VCdim($\mathcal{H}_{\sin}$)

To prove $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$, we need to pick n points which are shattered by $\mathcal{H}_{\sin}$, for any n . To do so, we construct n points $x_1, x_2, \dots, x_n \in [0, 1]$, such that the set of the m -th bits in the binary expansion, as m ranges from 1 to 2^n , ranges over all possible labelings of $x_1, x_2, \dots, x_n$.

$$\begin{array}{rcl} x_1 & = & 0.0 \mathbf{0} 0 0 \dots 1 1 \\ x_2 & = & 0.0 \mathbf{0} 0 0 0 \dots 1 1 \\ & \dots & \\ x_{n-1} & = & 0.0 \mathbf{0} 1 1 \dots 1 1 \\ x_n & = & 0.0 \mathbf{1} 0 1 \dots 0 1 \end{array}$$

$m=2$

If we wish to give the labeling 1 for $x_1, x_2, \dots, x_{n-1}$, and the labeling 0 for x_n , we pick $m=2$:

$$h(x) = \left\lceil \sin(2^2 \pi x) \right\rceil = 1 - \text{second_bit_of_binary_repres_of_} x$$

which returns 1 - the second bit (column) in the binary expansion.

VCdim($\mathcal{H}_{\sin}$)

To prove $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$, we need to pick n points which are shattered by $\mathcal{H}_{\sin}$, for any n . To do so, we construct n points $x_1, x_2, \dots, x_n \in [0, 1]$, such that the set of the m -th bits in the binary expansion, as m ranges from 1 to 2^n , ranges over all possible labelings of $x_1, x_2, \dots, x_n$.

$$\begin{array}{rcl} x_1 & = & 0.0000\dots 11 \\ x_2 & = & 0.0000\dots 11 \\ & \dots & \\ x_{n-1} & = & 0.0011\dots 11 \\ x_n & = & 0.0101\dots 01 \end{array}$$

$m=2^n$

If we wish to give the labeling 0 for $x_1, x_2, \dots, x_{n-1}, x_n$ we pick $m=2^n$:

$$h(x) = \left\lfloor \sin(2^{2^n} \pi x) \right\rfloor$$

which returns 1 - the bit on position 2^n (column) in the binary expansion.

$\text{VCdim}(\mathcal{H}_{\text{sin}})$

To prove $\text{VCdim}(\mathcal{H}_{\text{sin}}) = \infty$, we need to pick n points which are shattered by $\mathcal{H}_{\text{sin}}$, for any n . To do so, we construct n points $x_1, x_2, \dots, x_n \in [0, 1]$, such that the set of the m -th bits in the binary expansion, as m ranges from 1 to 2^n , ranges over all possible labelings of $x_1, x_2, \dots, x_n$.

$$x_1 = 0.0000\dots 11$$

$$x_2 = 0.0000\dots 11$$

...

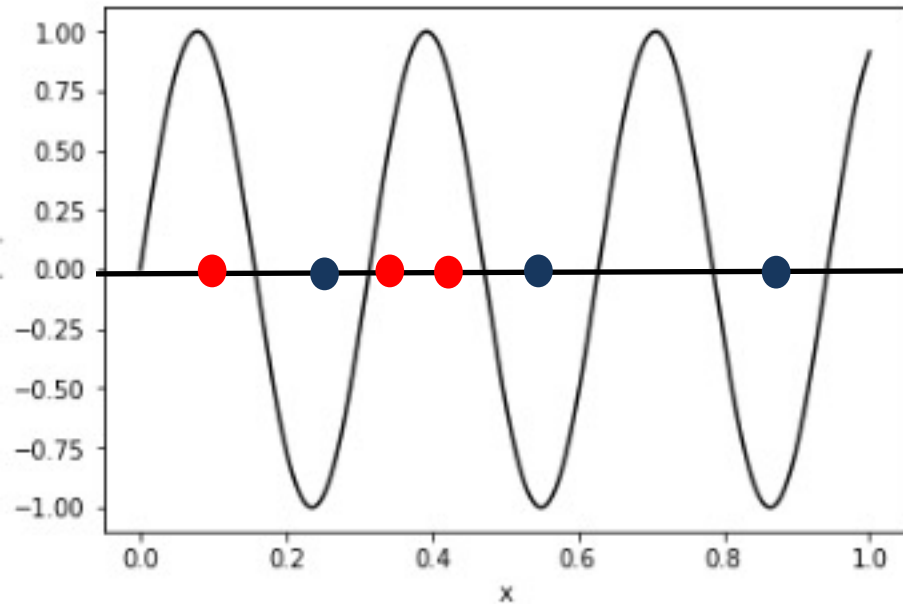
$$x_{n-1} = 0.0011\dots 11$$

$$x_n = 0.0101\dots 01$$

We conclude that $x_1, x_2, \dots, x_{n-1}, x_n$ can be given any labeling by some $h \in \mathcal{H}_{\text{sin}}$, so it is shattered. This can be done for any n , so $\text{VCdim}(\mathcal{H}_{\text{sin}}) = \infty$.

$\text{VCdim}(\mathcal{H}_{\sin})$

We conclude that $x_1, x_2, \dots, x_{n-1}, x_n$ can be given any labeling by some $h \in \mathcal{H}_{\sin}$, so it is shattered. This can be done for any n , so $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$.



Training points

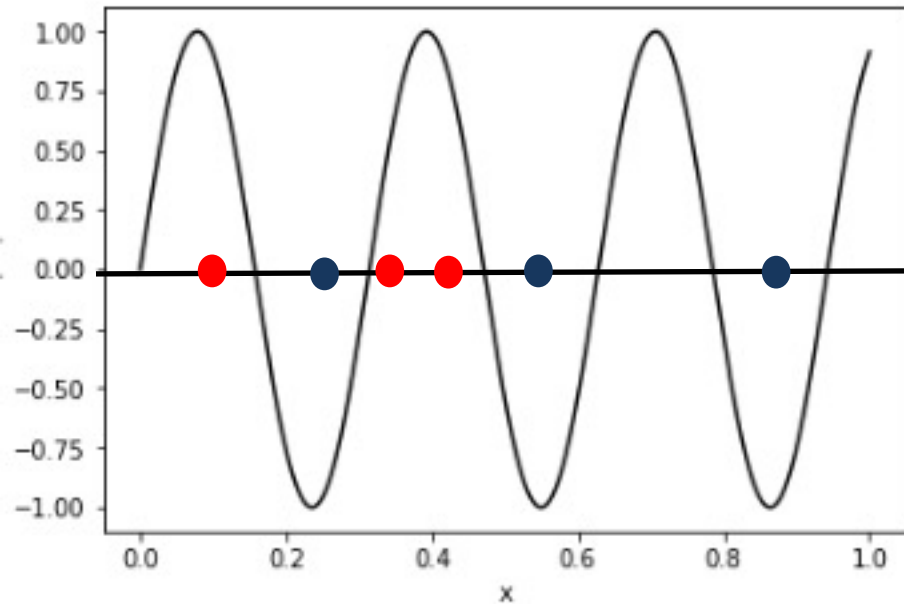
● Class 0

● Class 1

$$h(x) = \lceil \sin(20x) \rceil$$

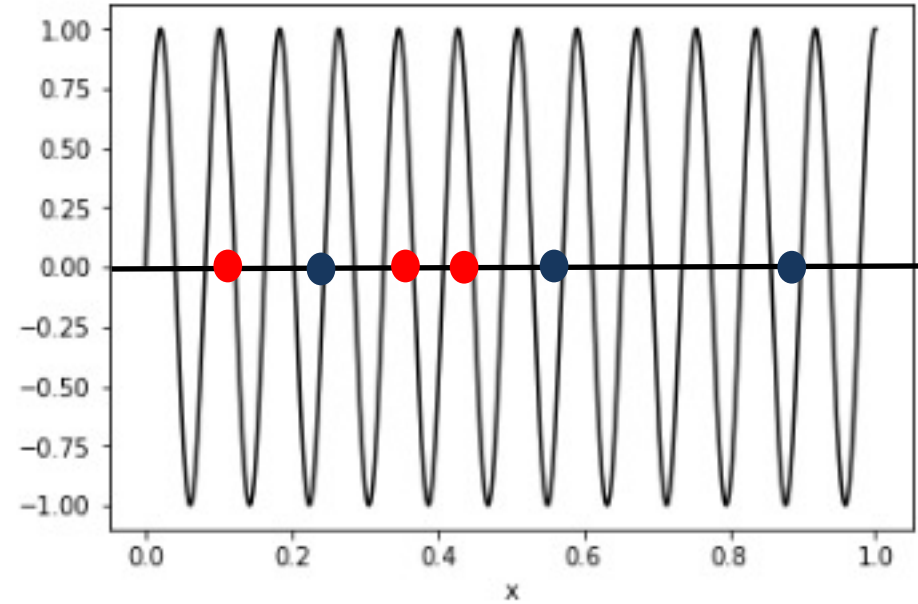
$\text{VCdim}(\mathcal{H}_{\sin})$

We conclude that $x_1, x_2, \dots, x_{n-1}, x_n$ can be given any labeling by some $h \in \mathcal{H}_{\sin}$, so it is shattered. This can be done for any n , so $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$.



- Class 0
- Class 1

$$h(x) = \lceil \sin(20x) \rceil$$

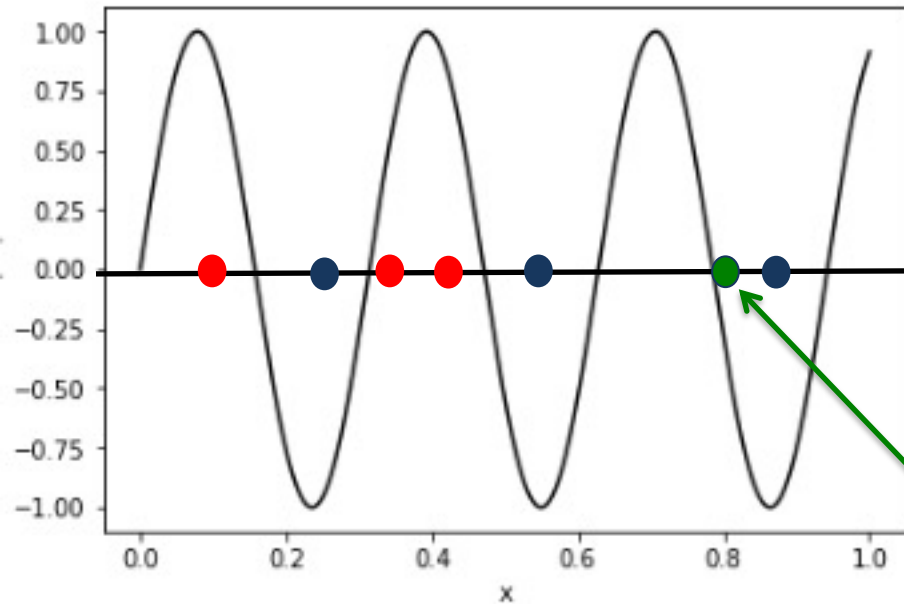


- Class 0
- Class 1

$$h(x) = \lceil \sin(77x) \rceil$$

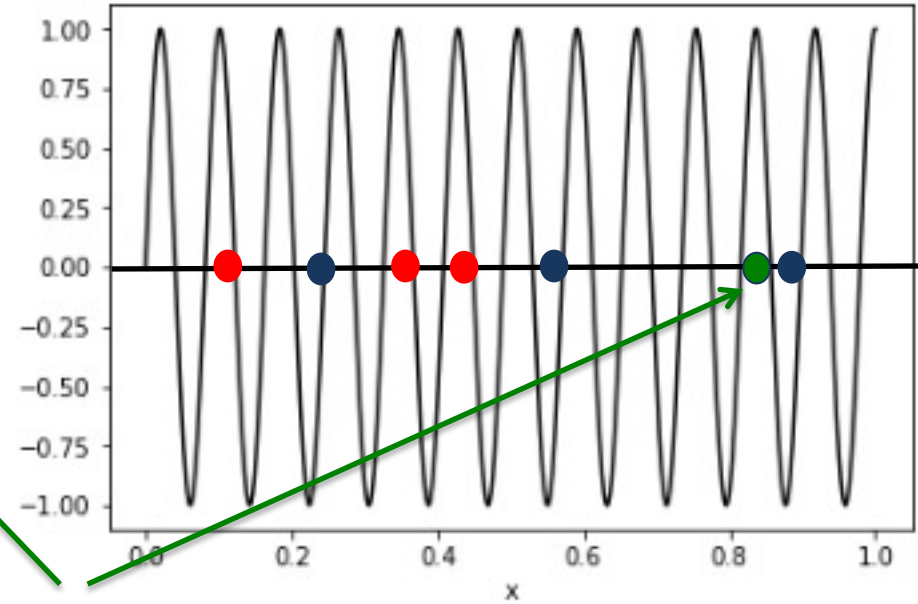
$\text{VCdim}(\mathcal{H}_{\sin})$

We conclude that $x_1, x_2, \dots, x_{n-1}, x_n$ can be given any labeling by some $h \in \mathcal{H}_{\sin}$, so it is shattered. This can be done for any n , so $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$.



- Class 0
- Class 1

$$h(x) = \lceil \sin(20x) \rceil$$

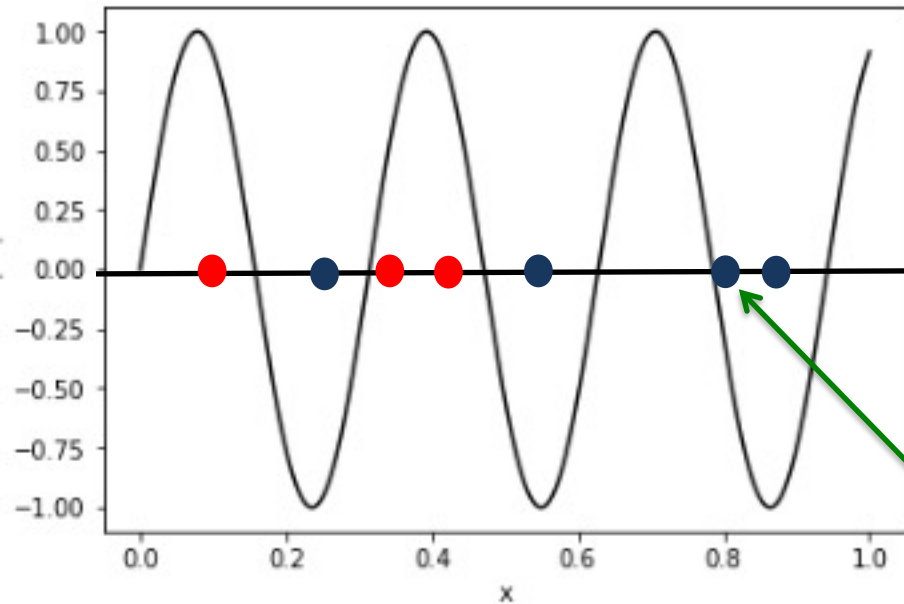


- Class 0
- Class 1

$$h(x) = \lceil \sin(77x) \rceil$$

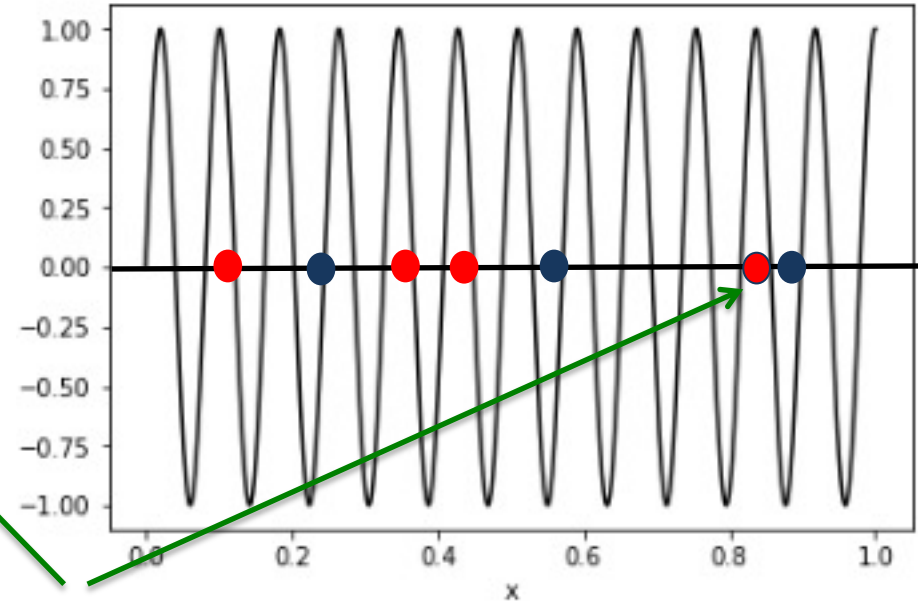
$\text{VCdim}(\mathcal{H}_{\sin})$

We conclude that $x_1, x_2, \dots, x_{n-1}, x_n$ can be given any labeling by some $h \in \mathcal{H}_{\sin}$, so it is shattered. This can be done for any n , so $\text{VCdim}(\mathcal{H}_{\sin}) = \infty$.



- Class 0
- Class 1

$$h(x) = \lceil \sin(20x) \rceil$$



- Class 0
- Class 1

$$h(x) = \lceil \sin(77x) \rceil$$